

Project Report: House Prices Prediction using Advanced Machine Learning

1. Introduction

This report presents a complete machine learning pipeline for predicting house prices using the Ames Housing dataset from the Kaggle 'House Prices: Advanced Regression Techniques' competition. The project focuses on feature engineering, model development, and ensemble learning to optimize prediction accuracy.

2. Dataset Description

The dataset contains 1,460 training rows and 1,459 test rows. It includes 79 explanatory variables covering structural, quality, neighborhood, and temporal properties of each house. Missing values, heterogeneous feature types, and non-linear interactions make this a challenging regression problem.

3. Exploratory Data Analysis

SalePrice is highly right-skewed, and log-transforming it yields a more symmetric distribution. Multiple features contain missing data, particularly quality-related attributes such as PoolQC, FireplaceQu, and Garage information.

4. Feature Engineering

Feature engineering significantly improves predictive performance. Key engineered features include TotalSF, TotalBath, TotalPorchSF, age-related attributes, interaction terms (e.g., GrLivArea $\times$ OverallQual), polynomial age features, and ordinal encodings for quality categories.

5. Modeling Approach

Several machine learning algorithms were trained, including Random Forest, XGBoost, LightGBM, and CatBoost. XGBoost with advanced features achieved strong validation performance. A final ensemble of four models further improved accuracy.

6. Ensemble Method

The final model blends XGBoost, LightGBM, CatBoost, and Random Forest using weighted averaging. This approach reduces variance and improves generalization, achieving the best Kaggle leaderboard score.

7. Results

The final blended model achieved a Kaggle leaderboard score of 13801, outperforming all single-model configurations. Feature engineering and ensembling had the largest impact on results.

8. Conclusion

This project demonstrates the importance of feature engineering and ensemble methods in tabular machine learning tasks. The pipeline is modular, interpretable, and competitive for this benchmark dataset.

9. Future Work

Potential improvements include K-fold cross-validation with out-of-fold predictions, Bayesian hyperparameter tuning, more domain-specific feature engineering, and advanced stacking models.